

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
Third Semester of B. Tech. Theory Examination (EC) Nov-Dec-2018
MA243.01 Transforms and Vector Differential Calculus

Date: 19/11/2018 (Monday) Time: 10:00 am to 01:00 pm Maximum Marks: 70

Instructions:

- i. The question paper contains two sections.
- ii. Section I and II must be attempted in separate answer sheets.
- iii. Figures to the right indicate marks.
- iv. Make suitable assumptions and draw neat figure where it is required.
- v. All notations and terminologies are standard.

Section-I

Q-1 Choose the correct answer from the given options in the followings: [07]

- 1) _____ is an even function of 't'.
 (A) t^2 (B) $t^2 - 4t$ (C) $\sin(2t + 3)$ (D) $t^3 + 6$
- 2) The Laplace transform of 1 is _____.
 (A) s (B) $\frac{1}{s}$ (C) $\frac{s}{2}$ (D) $\frac{2}{s}$
- 3) The value of $1 + \cos n\pi$ when $n = 7$ is _____.
 (A) 1 (B) 2 (C) 0 (D) None of these
- 4) The Laplace transform is _____.
 (A) Linear operation (B) Nonlinear operation (C) Static operation (D) Dynamic operation
- 5) A function f is said to be periodic with period T if _____.
 (A) $T = t$ (B) $f(t + T) = T$ (C) $f(t + T) = f(T)$ (D) $f(t + T) = f(t)$
- 6) The Laplace transform changes the domain of function into _____.
 (A) Z - domain (B) S - domain (C) R - domain (D) None of these
- 7) $L^{-1}\left\{\frac{1}{s^n}\right\}$ is possible when n is _____.
 (A) zero (B) negative integer (C) positive integer (D) negative rational

Q-2 Attempt any Four of the followings. [16]

- 1) Obtain the Fourier series for the function $f(x) = x^2$, $-\pi < x < \pi$. Hence show that

$$\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots = \frac{\pi^2}{6}.$$

- 2) Find the Laplace transform of following functions:

(i) $f(t) = t^2 \sin at$ (ii) $f(t) = \frac{\cos at - \cos bt}{t}$

3) Using Laplace transform, solve $\frac{d^2 y}{dt^2} + 4y = \sin t$ with $y(0) = 1$ and $y'(0) = 0$.

4) Obtain the Fourier series for the function,

$$f(x) = \begin{cases} 1 + \frac{2x}{\pi}, & -\pi \leq x \leq 0 \\ 1 - \frac{2x}{\pi}, & 0 \leq x \leq \pi \end{cases}.$$

5) Obtain Fourier series for $f(x) = e^{-x}$ in the interval $0 < x < 2\pi$.

6) Find inverse Laplace transform of following functions:

$$(i) \frac{se^{-2s}}{s^2 + \pi^2} \quad (ii) \frac{1}{(s + \sqrt{2})(s + \sqrt{3})}$$

Q-3 (a) Attempt any Two of the followings.

[06]

1) Express half range cosine series of $f(x) = x$ in the interval $(0, \pi)$.

2) Using convolution theorem, evaluate $L^{-1}\left[\frac{1}{s^2(s^2 + a^2)}\right]$.

3) Find the Laplace transform of $f(t)$ defined as

$$f(t) = \begin{cases} t, & 0 < t < 4 \\ 5, & t > 4 \end{cases}.$$

Q-3 (b) Attempt any One of the followings.

[06]

1) If $L[f(t)] = \bar{f}(s)$, then prove that $L[e^{at} f(t)] = \bar{f}(s - a)$. Hence evaluate

$$L\{e^{-2t}(4\cosh 3t - 2\sinh t)\}.$$

2) (i) Find the Fourier Transform of $e^{-a^2 x^2}$, $a < 0$.

$$(ii) \text{ Find the Fourier cosine transform of } f(x) = \begin{cases} x & , 0 < x < \frac{1}{2} \\ 1 - x & , \frac{1}{2} < x < 1 \\ 0 & , x > 1 \end{cases}.$$

Section-II

Q-4 Choose correct answer from the given options in the followings:

[07]

- 1) If $F(s)$ is the Fourier transform of $f(x)$ then $F[f(ax)]$ is _____.
 (A) $\frac{1}{a}F\left(\frac{s}{a}\right)$ (B) $\frac{1}{a}F(s)$ (C) $\frac{1}{s}F\left(\frac{s}{a}\right)$ (D) $aF\left(\frac{s}{a}\right)$
- 2) If $F(s)$ is the Fourier transform of $f(x)$ then $F[f(x-a)]$ is _____.
 (A) $e^{as}F(s)$ (B) $e^{-as}F(s)$ (C) $e^{ias}F(s)$ (D) $e^{-ias}F(s)$
- 3) If $u_n = \{15, 10, 7, \underset{\uparrow}{4}, 1, -1, 0, 3, 6\}$ then $u_3 =$ _____.
 (A) 0 (B) 15 (C) 3 (D) -1
- 4) The value of $Z\left(\sin \frac{n\pi}{2}\right)$ is _____.
 (A) $\frac{z^2}{z^2+1}$ (B) $\frac{z}{z^2+1}$ (C) $\frac{-z}{z^2+1}$ (D) None of these
- 5) A unit normal to $x^2 + y^2 + z^2 = 5$ at $(0, 1, 2)$ is equal to _____.
 (A) $\frac{\hat{i} + \hat{j} + \hat{k}}{\sqrt{5}}$ (B) $\frac{\hat{i} + \hat{j} - \hat{k}}{\sqrt{5}}$ (C) $\frac{\hat{j} + 2\hat{k}}{\sqrt{5}}$ (D) $\frac{\hat{i} - \hat{j} + \hat{k}}{\sqrt{5}}$
- 6) The angle between $\vec{a} = \hat{i}$ and $\vec{b} = \hat{j}$ is _____.
 (A) 0 (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{2}$ (D) π
- 7) There is no change in magnitude of a vector in $\vec{F}$, if _____.
 (A) $\nabla \times \vec{F} \neq 0$ (B) $\nabla \times \vec{F} = 0$ (C) $\text{div}(\vec{F}) \neq 0$ (D) $\text{div}(\vec{F}) = 0$

Q-5 Attempt any Four of the followings.

[16]

- 1) Find the Fourier sine and cosine transform of $f(x) = \begin{cases} 1, & 0 \leq x < 2 \\ 0, & x \geq 2 \end{cases}$.
- 2) (i) Find Z - transform of sequence $u_n = a^n$.
 (ii) If $u_n = \{8, 6, 3, -1, 0, 1, 4, 5\}$ then find $Z(u_n)$.
- 3) Find the value of $X(0)$ for
 (i) $X(z) = \frac{2z^2 + 3z + 12}{(z-1)^4}$ (ii) $X(z) = \frac{10z}{(z-1)(z-2)}$.
- 4) Find the constants a and b so that the surface $ax^2 - zby = (a+2)x$ will be orthogonal to the surface $4x^2y + z^3 = 4$ at point $(1, -1, 2)$.
- 5) Prove that $\vec{F} = (y^2 - z^2 + 3yz - 2x)\hat{i} + (3xz + 2xy)\hat{j} + (3xy - 2xz + 2z)\hat{k}$ is both solenoidal and irrotational.
- 6) If $\vec{V} = yz\hat{i} + xz\hat{j} + xy\hat{k}$, find a scalar function $u(x, y, z)$ such that $\vec{V} = \text{grad}(u)$.

Q-6 (a) Attempt any Two of the followings.

[06]

- 1) Define one sided Z transform. If $U_n = 1$ find the Z transform of U_{n-1} and $U_n - U_{n-1}$.
- 2) State convolution theorem for Fourier transforms. Find the Fourier cosine transform of $f(x) = 5e^{-3x} + 2e^{-2x}$.
- 3) Find the Fourier transform of $f(x) = \begin{cases} 1 - x^2, & |x| \leq 1 \\ 0, & |x| > 1 \end{cases}$. Hence deduce that $\int_0^\infty \frac{\sin t - t \cos t}{t^3} dt = \frac{\pi}{4}$.

Q-6 (b) Attempt any One of the followings.

[06]

- 1) Define gradient of a scalar field and curl of a vector field. Find the directional derivative of a scalar function $f(x, y, z) = x^2 + xy + z^2$ at the point $A(1, -1, -1)$ in the direction of the line AB, where B is $(3, 2, 1)$.
- 2) Find the Z transform of the followings:

(i) $\frac{1}{n!}(a^n + a^{-n})$ (ii) $\frac{1}{5}2^n - \frac{1}{5}(-3)^n - 5n(-3)^{n-1}$ (iii) $\{ {}^{k+n}C_n \}$
